



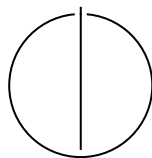
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Morphological Inflection Generation

Murilo Escher Pagotto Ronchi





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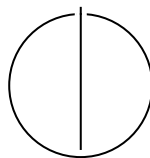
TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Morphological Inflection Generation

Titel der Abschlussarbeit

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Submission Date:	Submission date



I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, Submission date

Murilo Escher Pagotto Ronchi

Acknowledgments

Abstract

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1 Introduction

Citation test (Lamport, 1994).

Acronyms must be added in `main.tex` and are referenced using macros. The first occurrence is automatically replaced with the long version of the acronym, while all subsequent usages use the abbreviation.

E.g. `\ac{TUM}`, `\ac{TUM}` $\Rightarrow$ Technical University of Munich (TUM), TUM

For more details, see the documentation of the `acronym` package¹.

See Table 1.1, Figure 1.1, Figure 1.2, Figure 1.3.

Table 1.1: An example for a simple table.

A	B	C	D
1	2	1	2
2	3	2	3

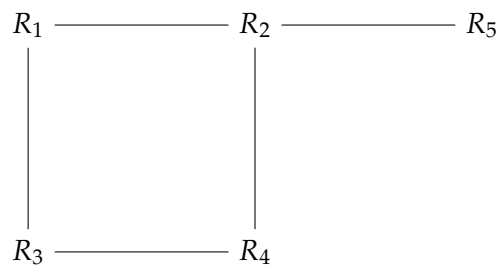


Figure 1.1: An example for a simple drawing.

¹<https://ctan.org/pkg/acronym>

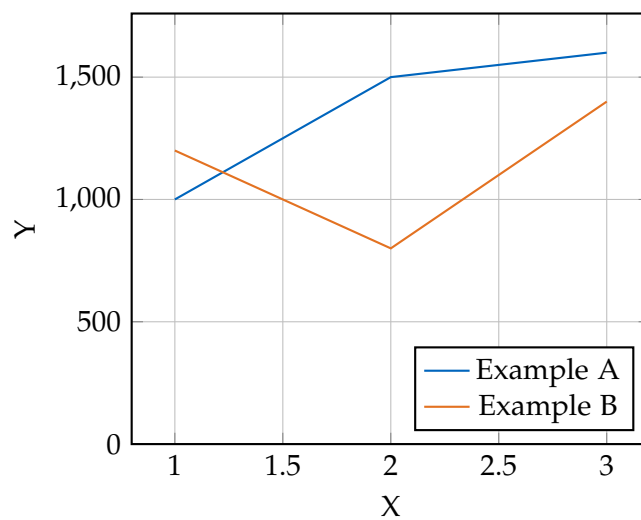


Figure 1.2: An example for a simple plot.

```
SELECT * FROM tbl WHERE tbl.str = "str"
```

Figure 1.3: An example for a source code listing.

2 Background

Different models have been used in this thesis to tackle two different, but related tasks. This chapter presents both tasks, the theoretical background and functionality behind the models used, as well as explanations of linguistic concepts pertaining to the tasks.

2.1 Tasks

This thesis addresses two closely related problems that can be viewed as inverse linguistic operations. Both tasks, introduced in the following subsections, involve applying a set of linguistic transformations to a base form of a word in order to produce an inflected form.

A lemma, commonly referred to as the root or canonical form, denotes the base representation of a word. This base form can undergo various transformations belonging to different grammatical categories, including mood, aspect, tense, case, number, and person. These transformations are represented as a collection of morphological features, which in this work are often referred to as *morphological tags*. Applying a specific set of such features to a lemma yields the corresponding inflected form.

2.1.1 Morphological Inflection

Morphological inflection refers to the process of generating a surface word form from a canonical base form (lemma) and a set of grammatical properties. Given a lemma and features describing tense, number, person, case, or other morphosyntactic categories, the task consists of producing the corresponding inflected form. An example in English is given in Figure 2.1.

$$(walk, \{\text{VERB, PAST, THIRD PERSON, SINGULAR}\}) \rightarrow walked$$

Figure 2.1: Illustration of the morphological inflection task. Given a lemma and a set of grammatical features, the goal is to generate the corresponding inflected form.

Automatic morphological inflection has long been studied in computational linguistics. Early approaches relied on rule-based systems and finite-state transducers (Koskenniemi, 1983; Kaplan and Kay, 1994; Beesley and Karttunen, 2003), which required curated rules determined by linguistic experts. More recently, the problem has been formulated as a supervised sequence-to-sequence learning task, in which neural encoder-decoder models operate directly on character sequences (Faruqui et al., 2016; Kann and Schütze, 2016; Aharoni and Goldberg, 2017). Such models learn morphological transformations from data and have become the dominant paradigm for the task.

The task has gained particular prominence through a series of multilingual shared tasks, most notably the SIGMORPHON and CoNLL–SIGMORPHON reinflection challenges (Cotterell, Kirov, Sylak-Glassman, Walther, Vylomova, Xia, et al., 2017; Cotterell, Kirov, Sylak-Glassman, Walther, Vylomova, McCarthy, et al., 2018; Goldman et al., 2023). These benchmarks provide standardized training and evaluation settings across dozens of typologically diverse languages and have established morphological inflection as a common testbed for studying character-level generalization and low-resource learning. Large-scale resources such as UniMorph further facilitate this research by offering broad cross-linguistic coverage and consistent morphological annotations (Batsuren et al., 2022).

2.1.2 Morphological Analysis

Full morphological analysis refers to the task of recovering both the lemma and the associated set of morphological descriptors from an inflected surface form. The task is closely related to the subtasks of lemmatization and morphological tagging, which predict the canonical form and grammatical features, respectively. An example in English is shown in Figure 2.2.

$$walked \rightarrow (walk, \{\text{VERB, PAST, THIRD PERSON, SINGULAR}\})$$

Figure 2.2: Illustration of the morphological analysis task. Given an inflected form, the goal is to predict the corresponding lemma and set of grammatical features.

Automatic morphological analysis has traditionally been addressed using rule-based and finite-state techniques similar to those employed for inflection. Classical two-level and finite-state morphology frameworks model analysis and generation within a single bidirectional system, enabling both the decomposition of surface forms into lexical representations and the production of inflected forms from abstract specifications

(Koskenniemi, 1983; Beesley and Karttunen, 2003).

More recently, data-driven approaches have largely replaced manually engineered analyzers. Related problems are often formulated as lemmatization or morphological tagging, where models predict either the lemma or the set of grammatical features directly from the surface form. Neural character-level architectures have proven effective for these tasks, learning morphological regularities directly from data without explicit rules (Malaviya et al., 2019; Cotterell and Heigold, 2017). In this thesis, we adopt the more general formulation of full morphological analysis, which jointly predicts both the lemma and the complete set of morphological descriptors.

Beyond these task formulations, accurate morphological modeling plays an important role in downstream natural language processing applications more broadly. Generating and analyzing correct surface forms is essential for systems that must produce grammatically well-formed text, such as machine translation and text generation, and for tasks that rely on reliable morphological features, including tagging and syntactic parsing. Prior work has shown that incorporating morphological or subword structure improves neural machine translation for morphologically rich languages (Sennrich et al., 2016), while morphology-aware representations and feature prediction benefit language modeling and tagging performance (Botha and Blunsom, 2014; Cotterell and Heigold, 2017). These findings motivate the study of both inflection and analysis as complementary components of robust morphological processing.

2.2 Inflection Baselines

2.2.1 Non-Neural Baseline

Earlier approaches to morphological inflection primarily relied on explicit hand-crafted transformations to generate inflected word forms. In line with this tradition, the first baseline evaluated in this work adopts a simple edit-rule strategy.

The system is provided as part of the SIGMORPHON 2023 Shared Task (Goldman et al., 2023). It builds upon a baseline introduced in the SIGMORPHON 2020 edition (Vylomova et al., 2020) and is based on the model originally proposed by Cotterell et al. (2017).

Unlike neural sequence-to-sequence approaches, this model does not learn continuous representations. Instead, it derives explicit string edit rules directly from the training data. For each training triple consisting of a lemma, a set of morphological tags, and the corresponding inflected form, the system first aligns the two strings using Levenshtein distance. The aligned pair is then decomposed into a prefix part, a shared stem, and a suffix part. Based on this decomposition, both prefix-changing and suffix-changing transformation rules are extracted.

To illustrate this procedure, we reproduce a simplified alignment example adapted from the official baseline documentation (Vylomova et al., 2020). For the German pair *schielen* $\rightarrow$ *geschielt*, one possible alignment is:

```
--schielen
geschielt_
```

which can be segmented into prefix (Pr), stem (St), and suffix (Su) regions:

```
Pr |   St   | Su
-- | schiele | n
ge | schielt | _
```

From this decomposition, the system extracts both prefix rules (e.g., $\epsilon \rightarrow ge$) and suffix rules describing the stem-final changes.

Each training example may therefore yield multiple candidate rules of varying specificity. During inference, the system first selects the rule whose left-hand side matches the longest portion of the lemma (suffixes for suffix rules or prefixes for prefix rules). If several rules of equal length apply, the most frequent one observed in the training data is chosen.

Table 2.1 illustrates representative examples of such transformations. For clarity, the examples focus on suffix edits, which are more common in English, although the same procedure is applied analogously to prefixes. By aggregating such examples across the training data, the model learns which modifications are most likely for a given morphological specification.

Importantly, each training pair contributes multiple candidate rules rather than a single transformation. For example, from the pair *walk* $\rightarrow$ *walked*, the alignment procedure yields a hierarchy of suffix edits with increasing specificity, including $\epsilon \rightarrow ed$, $k \rightarrow ked$, $lk \rightarrow lked$, $alk \rightarrow alked$, and *walk* $\rightarrow$ *walked*. This strategy encourages more specific transformations while still allowing general rules to apply when appropriate.

While this allows the model to efficiently capture regular patterns, it can lead to overgeneralization when training data is sparse. For instance, if the only observed pair were *speak* $\rightarrow$ *spoke*, the model could learn the rule *eak* $\rightarrow$ *oke*. Applying this rule to an unseen lemma such as *leak* would incorrectly yield *loke*, instead of *leaked*. This example illustrates how purely surface-based edits may fail to generalize appropriately.

Despite these limitations, the approach provides a simple, transparent, and computationally efficient baseline. Its explicit rule extraction makes the predictions easily interpretable and offers a useful reference point for comparing the more complex neural models introduced in the following sections.

Lemma	MSD	Inflected form	Example candidate rule
walk	V;PST;3;SG	walked	$\epsilon \rightarrow \text{ed}$
talk	V;PST;3;SG	talked	$\text{lk} \rightarrow \text{lked}$
play	V;PST;3;SG	played	$\text{ay} \rightarrow \text{ayed}$
jump	V;PST;3;SG	jumped	$\text{mp} \rightarrow \text{mped}$
call	V;PST;3;SG	called	$\text{all} \rightarrow \text{alled}$
love	V;PST;3;SG	loved	$\text{ve} \rightarrow \text{ved}$
move	V;PST;3;SG	moved	$\text{ove} \rightarrow \text{oved}$
try	V;PST;3;SG	tried	$\text{y} \rightarrow \text{ied}$
stop	V;PST;3;SG	stopped	$\text{p} \rightarrow \text{pped}$
fix	V;PRS;3;SG	fixes	$\epsilon \rightarrow \text{es}$

Table 2.1: Illustrative examples of edit rules extracted by the non-neural baseline. The table focuses on suffix rules for readability, although both prefix and suffix transformations are learned.

2.2.2 Neural Baseline

As a neural reference system, we use the official neural baseline provided as part of the SIGMORPHON 2023 Shared Task. This system corresponds to the character-level Transformer model proposed by Wu et al. (Wu et al., 2021), which has been adopted as a strong baseline for morphological inflection.

The model follows a standard encoder-decoder Transformer architecture (Vaswani et al., 2017), operating directly on character sequences. To better suit low-resource transduction tasks, a compact configuration is used, consisting of four encoder layers and four decoder layers with multi-head self-attention. Each layer employs four attention heads, a model dimension of $d_{\text{model}} = 256$, and a feed-forward dimension of $d_{\text{FF}} = 1024$. Decoding is performed autoregressively. For feature-guided tasks such as morphological inflection, morphological tags are encoded as special feature tokens that are concatenated with the input character sequence and processed jointly by the encoder. Wu et al. further propose a feature-invariant positional encoding scheme to avoid imposing an arbitrary order over the feature set.

2.3 ByT5

In addition to character-level and token-based models, we consider ByT5 (Xue, Barua, et al., 2022), a token-free variant of the T5 architecture. ByT5 is an encoder-decoder Transformer model that operates directly on UTF-8 byte sequences instead of subword

tokens. Unlike conventional Transformer models that rely on learned vocabularies (e.g., SentencePiece), ByT5 uses a fixed vocabulary of 256 byte embeddings corresponding to all possible byte values, thereby eliminating the need for tokenization.

Architecturally, ByT5 follows the T5 framework (Raffel et al., 2020) and its multilingual variant mT5 (Xue, Constant, et al., 2021). To better accommodate longer byte sequences, the encoder and decoder depths are decoupled, employing a substantially heavier encoder stack with a 3:1 encoder–decoder ratio, in contrast to the balanced configuration used in T5 and mT5.

Because ByT5 operates directly on raw byte sequences, it has been hypothesized to be particularly suitable for tasks that are sensitive to orthographic or phonological structure, including morphological inflection. This property makes it especially relevant for the evaluation conducted in this thesis.

In our experiments, ByT5 is fine-tuned on the target morphological task. The fine-tuning configuration is described in Section 4.

2.4 Large Language Models

Large language models (LLMs) are Transformer-based neural networks pretrained on large-scale text corpora using self-supervised objectives. Owing to their scale and broad pretraining, such models exhibit strong generalization capabilities and can perform a wide range of sequence-to-sequence tasks with little or no task-specific training. Since both morphological inflection and morphological analysis can be formulated as text generation problems, LLMs provide a natural evaluation framework without dedicated fine-tuning.

In this work, we evaluate two open-weight instruction-tuned LLMs: Qwen3-8B (Yang et al., 2025) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024). Both models are decoder-only Transformer architectures with approximately 8 billion parameters and are designed for general-purpose text generation. The 8B parameter scale represents a compromise between strong performance and practical computational requirements.

Qwen3-8B, developed by Alibaba, is part of the Qwen model family and is trained on large-scale multilingual data to provide strong general-purpose language understanding and generation capabilities. Llama-3.1-8B-Instruct, developed by Meta AI, belongs to the LLaMA series and incorporates improved pretraining data and alignment procedures relative to earlier versions. Both models are instruction-tuned to follow natural language prompts, making them suitable for prompt-based evaluation settings. In this thesis, they are used without additional fine-tuning and evaluated in a one-shot prompting setup.

3 Datasets and Languages

3.1 Datasets

To address the tasks of morphological inflection and morphological analysis and to train the proposed models, labeled data is required. Each instance consists of a lemma, a set of morphological tags describing the target grammatical features, and the corresponding inflected form. For this purpose, this work relies on two widely used resources: UniMorph 4.0 (Batsuren et al., 2022) and Universal Dependencies (UD) (Nivre et al., 2020).

3.1.1 UniMorph 4.0

UniMorph 4.0 is a multilingual project dedicated to providing large-scale morphological resources in the form of inflection tables and structured morphological annotations across a wide range of languages. At the time of writing, the dataset covers up to 182 languages, including both high-resource and low-resource as well as endangered languages.

For each language, UniMorph provides a dataset consisting of individual instances that pair a lemma with a bundle of morphological features and the corresponding inflected form. Concretely, the data is distributed as tab-separated lines, where each row represents a single lemma, tags, form triple. Multiple rows typically share the same lemma and correspond to different morphological feature combinations, effectively encoding the full inflectional paradigm in a flat, line-based format, as shown in Table 3.1.

Lemma	Tags	Inflected form
walk	V;PST;3;SG	walked

Table 3.1: Example instance of the tab-separated format used by UniMorph.

In this example, the morphological tags can be interpreted as shown in Table 3.2:

The UniMorph data served as the main resource throughout the experimental pipeline. Following the original SIGMORPHON shared task setup, it formed the

Tag	Meaning
V	Verb
PST	Past tense
3	Third person
SG	Singular

Table 3.2: Example UniMorph morphological tags and their meanings.

basis for the reimplementation of the shared task baselines.

Furthermore, the dataset was used to fine-tune the ByT5 model for morphological inflection and, by inverting the source–target mapping, also for morphological analysis. It additionally constituted the core training data for the bidirectional model, while certain experimental settings were further supplemented with data derived from Universal Dependencies.

3.1.2 Universal Dependencies

The UD resource provides consistent morphosyntactic annotation across languages, offering dependency treebanks with standardized part-of-speech tags, morphological features, and syntactic relations (Marneffe et al., 2021). At the time of writing, UD covers over 150 languages through more than 200 treebanks, spanning both high-resource and low-resource settings.

The data is distributed in the CoNLL-U format, where each sentence is represented as a sequence of token-level annotations. In addition to syntactic dependency information, each token is associated with a lemma and a set of morphological features encoded as attribute–value pairs (e.g., `Tense=Past|Number=Sing|Person=3`). Compared to UniMorph’s tab-separated triple format, UD therefore provides richer contextual and sentence-level information alongside morphological annotations. An example taken from the Portuguese PetroGold treebank is shown in Figure 3.1.

In this work, UD data serves as a source of contextual information for fine-tuning the bidirectional ByT5 model and for conducting experiments with large language models.

Since UD often provides multiple treebanks for a given language, it was necessary to select a single resource for each case. To guide this choice, we relied on the ranking provided by the UD project, which evaluates treebanks based on their size, annotation quality, and linguistic diversity. Whenever multiple treebanks were available for a language, the highest-ranked treebank according to this score was selected.

```
# text = Estas podem representar falhas profundas.
# sent_id = 247-20140910-MONOGRAFIA_0-42
1 Estas este PRON _ Gender=Fem|Number=Plur|PronType=Dem 2 nsubj _ _
2 podem poder VERB _ Mood=Ind|Number=Plur|Person=3|Tense=Pres|VerbForm=Fin
0 root _ _
3 representar representar VERB _ VerbForm=Inf 2 xcomp _ _
4 falhas falha NOUN _ Gender=Fem|Number=Plur 3 obj _ _
5 profundas profundo ADJ _ Gender=Fem|Number=Plur 4 amod _ SpaceAfter=No
6 . . PUNCT _ _ 2 punct _ _
```

Figure 3.1: Example sentence in CoNLL-U format from the Portuguese UD PetroGold treebank.

3.1.3 Data Preparation

Since the two data sources use different annotation schemes and file formats, several preprocessing and conversion steps were required to obtain a unified representation suitable for all experiments.

For UniMorph, we followed the official SIGMORPHON shared task setup whenever available, using the predefined training, development, and test splits provided for the corresponding languages. These partitions consist of 10,000 training instances, 1,000 development instances, and 1,000 test instances. For languages not covered by the shared task, the data was obtained directly from the original UniMorph repository and split into training, development, and test sets using the same size configuration to maintain consistency across languages.

For most UD treebanks, predefined training, development, and test splits are provided. However, for Amharic only a test set is available. In this case, the data was randomly partitioned into training, development, and test sets according to the same proportions as the other languages, as no official split specification is provided.

As the project builds upon a reimplementation of the SIGMORPHON baselines, the final data format closely follows the UniMorph specification. Each instance therefore consists of a lemma, a set of morphological tags, and the corresponding inflected form, separated by tab characters. For data originating from 3.1, an additional fourth column containing the sentence context is included when contextual information is required.

To incorporate UD data into this format, the morphological feature annotations provided in the CoNLL-U files were first mapped to their UniMorph equivalents. For this purpose, we used the official UniMorph conversion tool, which translates UD feature sets into the UniMorph schema using language-specific translators (McCarthy et

al., 2018). Since not all evaluated languages were supported by the provided translators, the conversion scripts were extended following the structure of the existing templates to additionally support Georgian and Amharic.

Figure 3.2 illustrates the output of the conversion tool when applied to the sentence shown in Figure 3.1. The UniMorph-compatible tags replace the original UD feature representations while preserving the original tokenization and syntactic structure.

```
# text = Estas podem representar falhas profundas.
# sent_id = 247-20140910-MONOGRAFIA_0-42
1 Estas este PRON _ PRO;PL;FEM 2 nsubj _ _
2 podem poder VERB _ PL;PRS;V;IND;3 0 root _ _
3 representar representar VERB _ V;NFIN 2 xcomp _ _
4 falhas falha NOUN _ PL;FEM;N 3 obj _ _
5 profundas profundo ADJ _ PL;FEM;ADJ 4 amod _ SpaceAfter=No
6 . . PUNCT _ _ 2 punct _ _
```

Figure 3.2: Example output from official conversion tool.

Since the UniMorph data predominantly contains verbs, UD data was filtered accordingly to include only verbal tokens. Concretely, for each sentence, the first verb was selected and used to construct a single instance.

After conversion, the lemma, the inflected form, the mapped UniMorph tags, and the surrounding sentence context were extracted to construct the final tab-separated instances used in the downstream experiments. The resulting unified format is illustrated in Table 3.3.

Lemma	Tags	Inflected form	Context
poder	V;IND;PRS;PL;3	podem	Estas podem representar falhas profundas.

Table 3.3: Example instance of the unified tab-separated format used in this work.

All preprocessing, conversion, and extraction scripts are included in the accompanying code repository to ensure full reproducibility.

3.2 Languages

The following tables summarize the languages considered in the experiments. Table 3.4 presents general typological and linguistic information, together with the UniMorph

data source for each language, indicating whether the data was obtained from the SIGMORPHON shared task or sampled directly from the original UniMorph repository. For all languages, UniMorph uses fixed training, development, and test splits of 10,000, 1,000, and 1,000 instances, respectively. In addition, the total number of UniMorph instances (“UM size”) is reported to reflect the overall amount of available training data. The average number of inflected forms per lemma is included as a coarse estimate of morphological richness. Finally, the linguistic diversity score proposed by (Joshi et al., 2020) is provided as an external measure of language resource availability.

Table 3.5 provides statistics for the Universal Dependencies resources. Since UD treebanks are not available for Amuzgo and Lower Sorbian, these languages are excluded from the contextual and large language model experiments. For the remaining languages, the selected treebank and the sizes of the training, development, and test splits are listed.

Language	Family	Type	Ling. score	UM size	Avg. forms
Amharic	Semitic	Templatic	2	46224	–
Amuzgo	Oto-Manguean	Polysynthetic	0	12204	–
Ancient Greek	Hellenic	Fusional	3	41593	–
Lower Sorbian	Slavic	Fusional	1	20121	–
English	Germanic	Analytic	5	115523	–
Italian	Romance	Fusional	4	509574	–
Georgian	Kartvelian	Agglutinative	3	74412	–
Portuguese	Romance	Fusional	4	303996	–

Table 3.4: Typological and UniMorph statistics for the selected languages.

Language	Treebank	Train	Dev	Test
Amharic	ATT	811	102	99
Ancient Greek	PTNK	692	424	404
English	GUM	7692	1220	1171
Italian	ISDT	10728	475	404
Georgian	GNC	798	120	741
Portuguese	PetroGold	5871	568	894

Table 3.5: Universal Dependencies treebanks and dataset sizes for the languages with available UD resources.

4 Methodology

4.1 Morphological Inflection

Placeholder text.

4.1.1 Non-Neural Baseline

Placeholder text.

As a reference system, we employ the non-neural baseline provided by the SIGMORPHON shared task repository. This model is rule-based and derives transformation rules directly from the training data, without relying on trainable neural parameters. Since the approach does not involve hyperparameter tuning or gradient-based optimization, it can be executed directly using the default configuration supplied by the original implementation.

For all experiments, the model is applied to the UniMorph data in the standard tab-separated format consisting of lemma, morphological tags, and target inflected form, as described in Section 3.

4.1.2 Neural Baseline

Placeholder text.

In addition to the rule-based system, we evaluate the neural baseline provided by the SIGMORPHON shared task. Unlike the non-neural approach, this model requires supervised training. The model is trained on the UniMorph training split and evaluated on the corresponding development and test sets. For languages not included in the official shared task configuration, the data is partitioned into training, development, and test sets as described in Section ??.

The model is trained using the hyperparameters specified in the original baseline implementation. The exact configuration used in our experiments is reported for reproducibility.

4.1.3 ByT5

Placeholder text.

For the morphological inflection task, we fine-tune the ByT5 model on the UniMorph training data. Each instance is formulated as a sequence-to-sequence problem in which the input consists of the lemma together with the corresponding morphological tags, and the target sequence corresponds to the inflected form. The model is evaluated on the held-out development and test splits described in Section ??.

Training is performed using PyTorch and the HuggingFace Transformers framework. All hyperparameters are fixed across languages unless stated otherwise. The complete training configuration is summarized in Figure ??.

4.1.4 Bidirectional Model

Placeholder text.

In addition to the task-specific ByT5 models, we train a single bidirectional model that jointly learns morphological inflection and morphological analysis within a unified sequence-to-sequence framework. For each language, the UniMorph training set is randomly partitioned into two equally sized subsets. One subset is used with the inflection mapping (lemma + morphological tags $\rightarrow$ inflected form), while the other subset is converted using the inverted mapping (inflected form $\rightarrow$ lemma + morphological descriptors). The resulting instances are combined to fine-tune a single shared model.

In the context of the morphological inflection experiments, this bidirectional model is evaluated by applying the inflection mapping to the test instances.

4.1.5 Evaluation

Placeholder text.

To evaluate the performance of the proposed models on the task of morphological inflection, we report several complementary metrics. The primary measure is exact-match accuracy, defined as the proportion of predicted inflected forms that exactly match the gold-standard forms:

In addition, we compute the average Levenshtein distance to quantify character-level deviations and provide a more fine-grained assessment of partial errors.

4.2 Morphological Analysis

Placeholder text.

4.2.1 ByT5

Placeholder text.

For morphological analysis, we adopt the same ByT5 architecture and training procedure, but reformulate the task by reversing the input–output mapping. In this setting, the inflected form serves as the input sequence, while the target output consists of the corresponding lemma and its morphological descriptors.

Apart from this change in data representation, the training setup and hyperparameters remain identical to those used for morphological inflection.

4.2.2 Bidirectional Model

Placeholder text.

We evaluate the same bidirectional model described in Section ?? . For morphological analysis, the model is applied using the reversed input–output mapping, where the inflected form is provided as input and the model predicts the lemma together with the corresponding morphological descriptors. Apart from the evaluation direction, the model, training data partitioning, and hyperparameter configuration are identical to the inflection setting.

4.2.3 Bidirectional Model with Context

Placeholder text.

4.2.4 Large Language Models

Placeholder text.

We evaluate large language models in an inference-only setting via the OpenAI API. Each instance is processed independently, i.e., every prediction is generated through a separate API request without retaining conversational history between examples. To reduce stochasticity and improve reproducibility, we fix the decoding configuration across all experiments by setting the temperature to 0.0 and by using a constant random seed.

We employ a one-shot prompting strategy. Each prompt consists of (i) a short instruction describing the task, (ii) a single labeled example drawn from the dataset, and (iii) the target instance for which a prediction is required. The same template and demonstration example are included in every request to ensure consistent behavior across instances. An example of the prompt template is shown in Figure ?? .

4.2.5 Evaluation

Placeholder text.

5 Results

5.1 Morphological Inflection

Placeholder text.

5.2 Morphological Analysis

Placeholder text.

6 Conclusion

Placeholder text.

Abbreviations

TUM Technical University of Munich

UD Universal Dependencies

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